

# The Hodgkin–Huxley membrane, in equations

ar006 · 3 July 2026

Almost all equations, but not quite: a few lines of prose between the blocks so the eye has somewhere to rest. The subject is the four-dimensional system for a space-clamped patch of excitable membrane, the model Hodgkin and Huxley fit to the squid giant axon in 1952 [1], with state  $\mathbf{x} = (V, m, h, n)^\top$ : one voltage and three gating variables. Everything below is that one system, taken apart one relation at a time.

## Current balance

The membrane is a capacitor in parallel with three voltage-gated conductances [1, 10]. Charge conservation across it gives the equation that ties the whole model together: the rate of change of voltage is set by the sum of ionic currents plus whatever is injected:

$$C_m \dot{V} = - \underbrace{\bar{g}_{\text{Na}} m^3 h (V - E_{\text{Na}})}_{I_{\text{Na}}} - \underbrace{\bar{g}_{\text{K}} n^4 (V - E_{\text{K}})}_{I_{\text{K}}} - \underbrace{g_{\text{L}} (V - E_{\text{L}})}_{I_{\text{L}}} + I_{\text{ext}}(t)$$

$C_m$ , membrane capacitance;  $\bar{g}_\bullet$ , peak conductances;  $E_\bullet$ , reversal potentials;  $m, h, n \in [0, 1]$ , gating variables;  $I_{\text{ext}}$ , injected current.

Each conductance is the peak value scaled by gates raised to a power: the exponents ( $m^3$ ,  $n^4$ ) are the number of independent subunits that must all be open at once. The gates are what make the patch excitable; the next few blocks say how they move.

## Gating kinetics

Each gate relaxes toward a voltage-dependent target, written two equivalent ways, as forward/backward rates, or as a target value  $x_\infty$  approached with time constant  $\tau_x$ :

$$\dot{x} = \alpha_x(V)(1 - x) - \beta_x(V)x = \frac{x_\infty(V) - x}{\tau_x}(V), \quad x_\infty = \frac{\alpha_x}{\alpha_x + \beta_x}, \quad \tau_x = \frac{1}{\alpha_x + \beta_x}, \quad x \in \{m, h, n\}.$$

$\alpha_x(V)$ ,  $\beta_x(V)$ , voltage-dependent opening/closing rate constants;  $x_\infty$ , steady-state (in)activation;  $\tau_x$ , its relaxation time constant;  $x \in \{m, h, n\}$ , the three gating variables.

## Rate functions ( $V$ in mV, rates in $\text{ms}^{-1}$ )

The rates themselves are the empirical heart of the model: six functions fit to voltage-clamp data, no first-principles derivation behind them [1]. They are what turns the abstract kinetics above into concrete dynamics:

$$\begin{aligned} \alpha_m(V) &= \frac{0.1(V + 40)}{1 - \exp(-(V + 40)/10)}, & \beta_m(V) &= 4 \exp(-(V + 65)/18), \\ \alpha_h(V) &= 0.07 \exp(-(V + 65)/20), & \beta_h(V) &= \frac{1}{1 + \exp(-(V + 35)/10)}, \\ \alpha_n(V) &= \frac{0.01(V + 55)}{1 - \exp(-(V + 55)/10)}, & \beta_n(V) &= 0.125 \exp(-(V + 65)/80). \end{aligned}$$

The  $\alpha_m$  and  $\alpha_n$  expressions look singular: numerator and denominator both vanish at one voltage, so a naive evaluation returns 0/0. The function is smooth there anyway: the singularity is removable, and the value is just the limit [9]:

$$\alpha_m(V) = \begin{cases} \frac{0.1(V+40)}{1 - \exp(-(V+40)/10)} & \text{if } V \neq -40 \\ 1 & \text{if } V = -40 \end{cases} \quad \text{since} \quad \lim_{V \rightarrow -40} \alpha_m(V) = 1.$$

## Steady-state gating as a Boltzmann sigmoid

Plotted against voltage, each  $x_\infty$  is an S-curve, and it is often more convenient to fit it directly as a Boltzmann sigmoid with a half-activation voltage  $V_{1/2}$  and a slope factor  $k$  than to go through the rate functions [7, 8]. Its derivative has the tidy logistic form that makes the slope at  $V_{1/2}$  exactly  $1/(4k)$ :

$$x_\infty(V) = \frac{1}{1 + \exp(-(V - V_{1/2})/k)}, \quad \frac{dx_\infty}{dV} = \frac{x_\infty(1 - x_\infty)}{k}.$$

$V_{1/2}$ , half-activation voltage;  $k$ , slope factor.

## Equilibrium potentials

The reversal potentials  $E_\bullet$  that appear in the current balance are not free parameters: they are set by the ionic concentrations either side of the membrane. Nernst gives the single-ion equilibrium; Goldman–Hodgkin–Katz generalises to the current carried by a partially permeant ion and to the resting potential of a membrane permeable to several ions at once [2, 3]:

$$E_S = \frac{RT}{z_S F} \ln \frac{[S]_{\text{out}}}{[S]_{\text{in}}}, \quad I_S = P_S z_S^2 \frac{VF^2}{RT} \cdot \frac{[S]_{\text{in}} - [S]_{\text{out}} e^{-z_S VF/RT}}{1 - e^{-z_S VF/RT}},$$

$$V_m = \frac{RT}{F} \ln \frac{P_K[K]_{\text{out}} + P_{\text{Na}}[\text{Na}]_{\text{out}} + P_{\text{Cl}}[\text{Cl}]_{\text{in}}}{P_K[K]_{\text{in}} + P_{\text{Na}}[\text{Na}]_{\text{in}} + P_{\text{Cl}}[\text{Cl}]_{\text{out}}}.$$

$E_S$ , Nernst potential of ion  $S$ ;  $V_m$ , resting potential of the mixed-permeability membrane;  $I_S$ , GHK current carried by  $S$ ;  $R$ , gas constant;  $T$ , absolute temperature;  $F$ , Faraday constant;  $z_S$ , valence of  $S$ ;  $P_\bullet$ , membrane permeabilities;  $[S]_{\text{in}}$ ,  $[S]_{\text{out}}$ , intra- and extracellular concentrations.

## Linear stability

With the vector field fully specified, the question of whether the rest state is quiet or poised to spike becomes a matter of eigenvalues [4, 5, 6]. Write the system as  $\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x})$  and linearise about a fixed point  $\mathbf{x}^*$  where  $\mathbf{F}(\mathbf{x}^*) = \mathbf{0}$ . The Jacobian  $\mathbf{J} = \partial \mathbf{F} / \partial \mathbf{x}$  is

$$\mathbf{J} = \begin{bmatrix} \partial_V \dot{V} & \partial_m \dot{V} & \partial_h \dot{V} & \partial_n \dot{V} \\ \partial_V \dot{m} & -1/\tau_m & 0 & 0 \\ \partial_V \dot{h} & 0 & -1/\tau_h & 0 \\ \partial_V \dot{n} & 0 & 0 & -1/\tau_n \end{bmatrix}, \quad \partial_V \dot{V} = -\frac{g_L + \bar{g}_{\text{Na}} m^3 h + \bar{g}_{\text{K}} n^4}{C_m},$$

and the rest state is stable iff every eigenvalue has negative real part:  $\det(\mathbf{J} - \lambda \mathbf{I}) = 0 \implies \text{Re}(\lambda_i) < 0$  for all  $i$ .

$\mathbf{F}$ , the model's vector field;  $\mathbf{x}^*$ , a fixed point;  $\mathbf{J}$ , the Jacobian there;  $\lambda_i$ , its eigenvalues;  $\partial_a \dot{b}$ , the partial of  $\dot{b}$  with respect to  $a$ .

## Synaptic drive and subthreshold response

So far  $I_{\text{ext}}$  has been an anonymous forcing term. In a network it is synaptic: presynaptic spikes open conductances that pull the membrane toward a synaptic reversal potential [7]. A spike train  $\{t_k\}$  arriving through a synapse of reversal  $E_{\text{syn}}$  adds

$$I_{\text{syn}}(t) = g_{\text{syn}}(t)(V - E_{\text{syn}}), \quad g_{\text{syn}}(t) = \bar{g} \sum_{k: t_k \leq t} \frac{t - t_k}{\tau_s} e^{1-(t-t_k)/\tau_s},$$

where each presynaptic spike contributes an alpha-function conductance transient of peak  $\bar{g}$  (the synaptic weight) that rises and decays on the synaptic time constant  $\tau_s$ . Below threshold, where

the membrane is a leaky integrator  $\tau \dot{V} = -(V - V_{\text{rest}}) + RI$  with membrane time constant  $\tau$  and input resistance  $R$ , the voltage is the convolution

$$V(t) = V_{\text{rest}} + \frac{R}{\tau} \int_{-\infty}^t e^{-(t-s)/\tau} I(s) \, ds, \quad \hat{Z}(\omega) = \frac{R}{1 + i\omega\tau},$$

with  $\hat{Z}(\omega)$  the input impedance, a first-order low-pass whose cutoff sits at  $\omega_c = 1/\tau$ .

$g_{\text{syn}}$ , synaptic conductance;  $E_{\text{syn}}$ , synaptic reversal potential;  $\{t_k\}$ , presynaptic spike times;  $\bar{g}$ , peak conductance (synaptic weight);  $\tau_s$ , synaptic time constant;  $\tau$ ,  $R$ , membrane time constant and input resistance;  $\hat{Z}(\omega)$ , input impedance;  $\omega_c$ , low-pass cutoff.

## References

1. Hodgkin AL, Huxley AF (1952). A quantitative description of membrane current and its application to conduction and excitation in nerve. *J Physiol* 117:500–544. doi:10.1113/jphysiol.1952.sp004764
2. Goldman DE (1943). Potential, impedance, and rectification in membranes. *J Gen Physiol* 27:37–60. doi:10.1085/jgp.27.1.37
3. Hodgkin AL, Katz B (1949). The effect of sodium ions on the electrical activity of the giant axon of the squid. *J Physiol* 108:37–77. doi:10.1113/jphysiol.1949.sp004310
4. FitzHugh R (1961). Impulses and physiological states in theoretical models of nerve membrane. *Biophys J* 1:445–466. doi:10.1016/S0006-3495(61)86902-6
5. Nagumo J, Arimoto S, Yoshizawa S (1962). An active pulse transmission line simulating nerve axon. *Proc IRE* 50:2061–2070. doi:10.1109/JRPROC.1962.288235
6. Ermentrout GB, Terman DH (2010). *Mathematical Foundations of Neuroscience*. Springer. doi:10.1007/978-0-387-87708-2
7. Gerstner W, Kistler WM, Naud R, Paninski L (2014). *Neuronal Dynamics*. Cambridge University Press. doi:10.1017/CBO9781107447615
8. Izhikevich EM (2007). *Dynamical Systems in Neuroscience*. MIT Press. doi:10.7551/mitpress/2526.001.0001
9. Sterratt D, Graham B, Gillies A, Willshaw D (2011). *Principles of Computational Modelling in Neuroscience*. Cambridge University Press. doi:10.1017/CBO9780511975899
10. Cole KS, Curtis HJ (1939). Electric impedance of the squid giant axon during activity. *J Gen Physiol* 22:649–670. doi:10.1085/jgp.22.5.649